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**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
Off-Campus Centre of Nitte (Deemed to be University)  
**Third Semester MCA (CBCS) Degree Examinations**

December 2025

**22MCA352 – NATURAL LANGUAGE PROCESSING**

Duration: 3 Hours

Max. Marks: 100

**Note: Answer Five full questions.**

| Unit – I  |                                                                                                                                                                                                                                                                                                                                                                                                  | Marks | BT* | CO* | PO* |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a)     | Explain the concept of Natural Language Processing. Describe the main focus of different disciplines involved in the study of language and the typical tools or techniques they use to solve language-related problems.                                                                                                                                                                          | 8     | L2  | 1   | 1   |
| b)        | Explain the difference between text-based applications and dialogue-based applications in NLP, providing examples of each.                                                                                                                                                                                                                                                                       | 6     | L2  | 1   | 1   |
| c)        | Explain an adverbial phrase and provide an example to illustrate its use in a sentence.                                                                                                                                                                                                                                                                                                          | 6     | L2  | 1   | 1   |
| 2. a)     | Explain the types of language knowledge and give an example for each. Consider the following sentence: "Language is one of the fundamental aspects of human behavior and is a crucial component of our lives." Analyze sentence with respect to syntactic, semantic, and pragmatic correctness, and explain which levels of language analysis are violated in the sentences.                     | 10    | L4  | 1   | 2   |
| b)        | Explain the elements of simple noun phrases and verb phrases with examples. Identify noun phrases and verb phrases in the following sentences.<br>i. The man who gave Bill the money...<br>ii. The rug that George gave to Ernest...<br>iii. The man whom George gave the money to ...                                                                                                           | 10    | L3  | 1   | 1   |
| Unit – II |                                                                                                                                                                                                                                                                                                                                                                                                  |       |     |     |     |
| 3. a)     | Explain the concept of the minimum edit distance algorithm. Using dynamic programming and backtracking, calculate the minimum edit distance required to transform the word "intention" into "execution".                                                                                                                                                                                         | 10    | L2  | 2   | 1   |
| b)        | Explain the concept of N-gram language modeling. Discuss bigram, trigram, and 4-gram probabilities with suitable examples to illustrate how they are computed and applied in predicting word sequences.                                                                                                                                                                                          | 10    | L3  | 2   | 2   |
| 4. a)     | Explain Part-of-speech tagging with example.                                                                                                                                                                                                                                                                                                                                                     | 10    | L2  | 2   | 1   |
| b)        | Given Corpus:<br>i. <S> I am Henry </S><br>ii. <S> I like college </S><br>iii. <S> Henry I am </S><br>iv. <S> I Do like Henry </S><br>v. <S> I do like Henry </S><br>Using the above corpus, compute the following n-gram probabilities.<br>i. <S> Do? (bi – gram probability)<br>ii. <S> I like Henry? (bi – gram)<br>iii. <S> DO I like ? (tri – gram)<br>iv. <S> Do I like college ( 4- gram) | 10    | L3  | 2   | 1   |

P.T.O.

## Unit – III

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|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 5. a) | Explain the concept of top-down parsing in syntactic analysis. Illustrate its working with a suitable example, showing how the parse tree is constructed from the start symbol to the input string. | 10 | L2 | 3 | 1 |
| b)    | Explain what context-free grammars are and discuss their significance in natural language processing. Provide examples to show how CFGs are used to represent the syntactic structure of sentences. | 10 | L3 | 3 | 2 |
| 6. a) | Define Transition Network Grammars and explain how they are used in parsing                                                                                                                         | 10 | L2 | 3 | 1 |
| b)    | Explain top-down chart parser with example.                                                                                                                                                         | 10 | L2 | 3 | 1 |

## Unit – IV

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|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 7. a) | Explain feature system in linguistics. Discuss how feature systems are applied in natural language parsing to analyze and represent linguistic information.             | 10 | L2 | 4 | 1 |
| b)    | Explain the concept of binary features with examples. Discuss how these features are used in linguistic analysis to distinguish between different elements of language. | 10 | L2 | 4 | 1 |
| 8. a) | Explain morphological analysis process and its importance in the lexicon. Give an example of morphological analysis for the word "Unwanted"                             | 10 | L2 | 4 | 1 |
| b)    | Describe a simple grammar using features and explain how it augments traditional parsing techniques.                                                                    | 10 | L2 | 4 | 1 |

## Unit – V

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|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|
| 9. a)  | Explain the concept of Named Entity Recognition (NER) and discuss its importance in information extraction. Illustrate with an example how NER can identify entities in a sentence.                        | 10 | L2 | 5 | 1 |
| b)     | Explain the process of relationship detection and classification in information extraction. Provide an example showing how relationships between entities in a sentence can be identified and categorized. | 10 | L2 | 5 | 1 |
| 10. a) | Describe the key techniques involved in temporal and event processing in information extraction applications.                                                                                              | 10 | L2 | 5 | 1 |
| b)     | Discuss how template filling is used in information extraction tasks. Provide an example.                                                                                                                  | 10 | L2 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome.

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